

Intentional Binding Experiment
PSY310 - Lab in Psychology
Lab Report

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GitHub Link:

https://github.com/aryanm4/PSY310_LabInPsychology/tree/Intentional-Binding

Introduction

Intentional binding is commonly seen as an implicit measure of agency since it implies a pre-reflective reduction of the perceived time between a deliberate act and its outcome, without necessarily stating that the act is one of control and causality. In normal paradigms the judged time of a voluntary keypress will shift later and the resulting tone will be perceived earlier, which causes an action and result to be more subjectively timed; the tendency of voluntary more than passive movements to do so is also reliably stronger.¹ The measure does not require direct self-attributions as it uses perceptual timing judgments, which reduces demand characteristics and self-report biases, consistent with other factors that agency may be implicitly sensorimotor-based in nature and anticipates reflective appraisal.

The results are convergent; outcome delays decrease both explicit agency rating and binding, and these two variables are correlated among individuals, which suggests that implicit and explicit agency processes across modalities are linked to each other. Clinical and mechanistic reviews also highlight its usefulness: changes in binding track disruption to predictive motor events involved in agency disruptions, it may be used as a sensitive, indirect measure of agency as opposed to a direct measure of agency itself.² Meanwhile, modern employment warns that predictability, causal beliefs, and task structure can be gated to bind; thus, it is only well understood in the context of explicit ratings and corroborating tasks to convey the multifariousness of agency.³ In general, intentional binding offers a strong, indirect view of how sensorimotor predictions anchor actions in terms of their effects in subjective time and makes it suitable as a common measure of the sense of agency.

¹ Cavazzana et al., 2014

² Desantis et al., 2012

³ Imaizumi & Tanno, 2018

Method

Participants

The conducted experiment had 1 participant. He was a male undergraduate student at Ahmedabad University, aged 18. He was fully informed of the experiment's objective before it was conducted and full consent was provided.

Materials and Procedure

Repetition of trials is achieved with three successive screens of action, tone and report timing with a short timing that is strictly regulated on the black background.

Action screen (trial): During the action screen, it will be shown as a white filled circle (Polygon) at the center of the screen and 1.0 s later this circle will be replaced by a magenta/pink circle (Polygon2) at the same location, after this time the participant is able to press the spacebar to perform their action. This is because the spacebar is activated as soon as the colored (magenta) circle is displayed (at 1.0 s), and the trial screen is terminated as soon as the participant presses a key; otherwise the routine will perform another loop until a response is received or the loop logic times out.

Tone screen (beep): When the action screen ends, the participants are presented with a blank display, and a brief tone (0.3 s) is sounded over the speakers, the frequency of which is defined by the trial conditions (freq), and the tone starts after a preset delay (delay) to the start of the beep routine, which enforces the action-outcome interval to be judged. The tone timing is similar to the display flip and will cease automatically after 0.3 s, after that the routine will cease and will continue to the report screen.

Report screen (response): On the top of the page there is a text prompt: Report the delay between the pressed key and the beep sound (0-999); Click submit after entering the value with an editable central textbox and a submit button placed below it. The participants enter an estimate of the delay between actions and tones (milliseconds) in the textbox and press the Submit button to proceed; they are prevented by the Submit button from continuing the routine, and the value typed in the textbox is stored with the timing of the click recorded before the next randomized trial commences.

Throughout the experiment, there are trials randomized to an external conditions file (ib_cons.xlsx) and repeated 5 times to the loop settings maintaining the action to tone to report sequence.

Results

Expected Condition: 276.768 seconds

Unexpected Condition 285.891 seconds

Difference in Interval Estimates -9.123

Discussion

The implicit measures such as purposeful binding minimize the demand properties and self-report bias that pollute direct agency measures. The fact that binding is performed at the perceptual level and does not necessitate any explicit realization of what is being measured means that participants can not easily respond in a particular way in accordance with social desirability or experimental anticipations. It is quantifiable and objective, dependent on perceptual timing decisions, not on subjective affects, and can be accurately and reproducibly quantified between individuals and situations. Also the implicit binding is a process that exploits pre-reflective sensorimotor processing, which occurs prior to higher-order conceptual verdicts, that offers an insight into the underlying agency mechanisms prior to the additional elaboration at the conscious level.

One important weakness is that implicit and explicit measures are not always correlated indicating that they are measuring dissimilar constructs and not a coherent agency experience. Empirical challenges of the recent past have raised questions as to whether intentional binding is indeed an index of intention; some research studies show that binding is not influenced by voluntary action in any way but by predictability and sensory integration. It is also very context-dependent and modifiable by task structure, outcome delays, and causal beliefs which makes it hard to generalize. Lastly, there is continued debate on whether a proxy of agency is valid through the use of binding especially when the latter develops without an evident intention. Therefore, implicit measures are beneficial in minimizing bias, but they might fail to represent the sheer amount of subjective agency and should be used in conjunction with explicit measures to conduct a comprehensive evaluation.

References

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